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QUASI-NORM LOCAL ERROR ESTIMATORS FOR p-LAPLACIAN*

WENBIN LIU† AND NINGNING YAN‡

Abstract. In this paper, we extend the quasi-norm techniques used in a priori error estimation of finite element approximation of degenerate nonlinear systems in order to carry out an improved a posteriori error analysis for the p-Laplacian. We derive quasi-norm a posteriori error estimators of residual type, which are shown to provide improved upper and lower bounds on the discretization error. For sufficiently regular solutions, these estimators are further shown to be equivalent on the discretization error in a quasi norm. Numerical results demonstrating these a posteriori estimators are also presented.

Key words. finite element approximation, p-Laplacian, a posteriori error estimators, quasi-norm error bounds

AMS subject classifications. 65N30, 49J40

PII. S0036142999351613

1. Introduction.

In this work we present an error analysis for the continuous piecewise linear finite element approximation of the p-Laplacian with Dirichlet data:

$$\begin{aligned} -\mathrm{div}(|\nabla u|^{p-2}\nabla u) &= f \quad \text{in } \Omega, \\ u &= g \quad \text{on } \partial\Omega, \end{aligned} \tag{1.1}$$

where $1 < p < \infty$ and Ω is a bounded open subset of R^n (n=1, 2, or 3) with a Lipschitz boundary $\partial\Omega$. Assumptions on the data f and g will be specified later. This equation is viewed as one of the classic examples of a large class of difficult problems—degenerate nonlinear systems. Indeed, it is believed that this simpler equation contains most of the essential difficulties in the studies of finite element approximation of this class of degenerate nonlinear systems, where many existing techniques (such as linearization or deformation procedure) in finite element method do not seem to work well. In fact, studies of the p-Laplacian have led to many active areas of research worldwide on the regularity and numerical approximation of the solutions of the degenerate nonlinear systems; see [13] and [11] for some previous work, and see [8] for an overview of some recent work. In [8] among other things, the quasi-norm approach, which has proved successful in deriving sharper a priori error bounds for finite element approximation of the degenerate systems, was summarized. Some very recent work on the p-Laplacian can be found in [12] for a mixed finite element method, in [2] and [3] for the h-p version of finite element method, and in [26] for a domain decomposition method and a multigrid method. Another important area is a posteriori error estimation of the p-Laplacian. The work in this area seems to date back to [22], and some recent work can be found in [6], [23], [8], and [29], where a posteriori error estimates of residual type were derived with both upper and lower bounds on the discretization error. However, in all the cases there are *gaps* in the power between

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the existing upper and lower estimates. For instance, it is can be shown from [6], [23], and [25] that there are positive constants, c, C such that for $1 < p \leq 2$

$$c|u - u_h|_{W^{1,p}(\Omega)} \leq \hat{\eta}_1 + \hat{\eta}_2 \leq C|u - u_h|^{p-1}_{W^{1,p}(\Omega)}$$

and for $p \geq 2$

$$c|u - u_h|_{W^{1,p}(\Omega)} \leq \left(\hat{\eta}_1^{p'} + \hat{\eta}_2^{p'}\right)^{1/p} \leq C|u - u_h|^{1/(p-1)}_{W^{1,p}(\Omega)},$$

where u_h is the piecewise linear finite element approximation of (1.1), and the posteriori error estimators $\hat{\eta}_1, \hat{\eta}_2$ were defined in [6]; see Remark 4.1 below. It does not seem to us that these gaps can possibly be removed by applying the standard residual techniques. With such a power gap, it is questionable whether or not the error indicators based on these estimates can adequately provide information on the approximation error and lead to efficient mesh refinement in adaptive approximation schemes. In this work we extend the quasi-norm techniques in order to derive some improved a posteriori error estimates for conforming finite element approximation of the p-Laplacian. The extension is certainly not trivial and seems to be applicable to more general degenerate nonlinear equations, e.g., those studied in [19]. In [20], some nonconforming finite element approximations of the p-Laplacian have also been examined.

In this paper we derive quasi-norm a posteriori error estimators with both lower and upper bounds on the discretization error of the piecewise linear finite element approximation of (1.1). It appears that there are substantial differences between the new estimators and the existing ones in their structures and the techniques used in deriving them. It seems that the structure of the new estimators reflects the degenerate nonlinear features of the p-Laplacian. For sufficiently regular solutions, these quasi-norm a posteriori error estimators are shown to be equivalent to the discretization error in a quasi norm. Initial analysis and numerical experiments indicate that the new estimators are sharper than the existing ones in some notable cases. Our numerical tests further suggest that the error indicators based on these quasi-norm a posteriori estimates can lead to more efficient mesh refinement schemes in some important cases.

The plan of this paper is as follows. In section 2, we give the p-Laplacian a variational formulation. Then we set up the continuous piecewise linear finite element approximation for the equation. We also introduce some quasi norms and related results. In section 3, we derive quasi-norm a posteriori upper and lower error bounds. In section 4, the main results in section 3 are further extended, and results of some numerical experiments are presented. Some important inequalities are included in the appendix.

Let Ω be a bounded open set in R^n $(n \leq 3)$ with a Lipschitz boundary $\partial\Omega$. In this paper we adopt the standard notation $W^{m,q}(\Omega)$ for Sobolev spaces on Ω with norm $\|\cdot\|_{m,q,\Omega}$ and seminorm $|\cdot|_{m,q,\Omega}$. We set $W_0^{m,q}(\Omega) \equiv \{w \in W^{m,q}(\Omega) : w|_{\partial\Omega} = 0\}$. We denote $W^{m,2}(\Omega)$ by $H^m(\Omega)$ with norm $\|\cdot\|_{m,\Omega}$ and seminorm $|\cdot|_{m,\Omega}$. In addition, c or C denotes a general positive constant independent of h.

2. Finite element approximation. In this section we study the weak formulation of the p-Laplacian and its finite element approximation. To this end, we need some important inequalities which are among the key ingredients of our a posteriori error estimation for finite element approximation of the p-Laplacian. These inequalities are included in the appendix.

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We make the following assumption on the data:

$$f \in L^2(\Omega), g \in W^{1-1/p,p}(\partial\Omega).$$

The weak formulation of the p-Laplacian is then as follows. (WP) For given f, g, find $u \in W_g^{1,p} = \{v \in W^{1,p}(\Omega) : (v-g)|_{\partial\Omega} = 0\}$ such that

$$a(u,v) = (f,v) \quad \forall v \in W_0^{1,p}, \tag{2.1}$$

where

$$a(u,v) = \int_\Omega |\nabla u|^{p-2}\nabla u \cdot \nabla v$$

and

$$(f,v) = \int_\Omega fv.$$

It is a simple matter to show that there exists a unique solution to (WP). There has been a great deal of work on the regularity of the solution u to (WP). For instance, for sufficiently regular data, global $C^{1,\alpha}$ regularity is established in [15]. Higher order regularity is investigated in [16] and [17]. In the rest of the paper, we assume that there is a $q > n$ such that $u \in W^{1,q}(\Omega)$. Consequently, the solution u is continuous.

We now consider the continuous piecewise linear finite element approximation of (WP). We also introduce some quasi norms which naturally arise in degenerate problems of this type and have proved useful in deriving sharper a priori error bounds for the finite element approximation of such problems; see, e.g., [7], [8], and [19].

Let Ω^h be a polygonal approximation to Ω with the boundary $\partial\Omega^h$. Let T^h be a partitioning of Ω^h into disjoint open regular simplices K, so that $\bar{\Omega}^h = \bigcup_{K\in T^h} \bar{K}$. Each element has at most one face on $\partial\Omega^h$, and $\bar{K}$ and $\bar{K}'$ have either only one common vertex, or a whole edge, or a whole face if K and $K' \in T^h$. We further require that $P_i \in \partial\Omega^h \Rightarrow P_i \in \partial\Omega$, where $\{P_i\}(i = 1 \ldots J)$ is the vertex set associated with the partitioning T^h. Let h_K denote the maximum diameter of the element K in T^h, and let ρ_K denote the diameter of the largest ball contained in K. We assume that there is a regularity constant R of T^h such that $1 \le \max_{K\in T^h}(h_K/\rho_K) \le R$.

We shall only discuss the continuous piecewise linear elements due to the limited higher order regularity for the solution of the p-Laplacian; see, for instance, [16] and [17], for the details.

Associated with T^h is a finite dimensional subspace V^h of $C^0(\bar{\Omega}^h)$, such that $\chi|_K$ are affine functions $\forall\ \chi \in V^h$ and $K \in T^h$. For ease of exposition we will assume that $\Omega^h = \Omega$, though all the results can be extended to the more general case where $\Omega^h \subset \Omega$.

Let

$$V_0^h = \{\chi \in V^h : \chi(P_i) = 0 \ \ \forall P_i \in \partial\Omega\},$$

$$V_g^h = \{\chi \in V^h : \chi(P_i) = g(P_i) \ \ \forall P_i \in \partial\Omega\}.$$

Let π_h: $C^0(\bar{\Omega}^h) \to V^h$ denote the standard piecewise Lagrange interpolation operator such that for any $v \in C^0(\bar{\Omega}^h)$, $\pi_h v(P_i) = v(P_i), i = 1 \to J$.

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Then, the finite element approximation of (WP) is as follows. Find a $u_h \in V_g^h$ such that

$$(2.2)\quad (WP)^h \quad a(u_h, v_h) = \int_{\Omega^h} |\nabla u_h|^{p-2} \nabla u_h \cdot \nabla v_h = \int_{\Omega^h} f v_h = (f, v_h) \ \ \forall v_h \in V_0^h.$$

It is also a simple matter to show that $(WP)^h$ has a unique solution u_h. Analysis of the finite element approximation of (WP) and a priori error bounds for this approximation are discussed in [11], [13], and [10]. Some of the best results are obtained in [7].

One of the key ideas in our approach is to introduce some quasi norms to handle the possible degeneracy of the p-Laplacian in order to obtain the sharpest error bounds. As will be seen later, this idea is also essential in deriving sharper a posteriori error estimates for the p-Laplacian. In the following we very briefly introduce these quasi norms and some relations between them and the standard Sobolev norms.

Let $w \in W^{1,p}(\Omega)$. Let $\rho = \max\{2, p\}$. As in [9], we define for any $v \in W^{1,p}(\Omega)$

$$(2.3)\qquad \|v\|_{(w,p)}^{\rho} \equiv \int_{\Omega} |\nabla v|^2 (|\nabla w| + |\nabla v|)^{(p-2)}.$$

It has been shown in [9] that $\|\cdot\|_{(w,p)}$ is a quasi norm on $W_0^{1,p}(\Omega)$: more precisely the following has been shown.

PROPOSITION 2.1. *The function $v \to \|v\|_{(w,p)}$ is a quasi norm on $W_0^{1,p}(\Omega)$, that is,*

(i) $\|v\|_{(w,p)} \geq 0$ *and* $\|v\|_{(w,p)} = 0$ *if and only if* $v = 0$;

(ii) $\|v_1 + v_2\|_{(w,p)} \leq \|v_1\|_{(w,p)} + \|v_2\|_{(w,p)}$ *for any* $v_1, v_2 \in W^{1,p}(\Omega)$.

Consequently, $d(v_1, v_2) = \|v_1 - v_2\|_{(w,p)}$ is a distant on $W_0^{1,p}(\Omega)$.

Furthermore, for $1 < p \leq 2$, we have

$$(2.4)\qquad |v|_{W^{1,p}(\Omega)} \leq C(|w|_{W^{1,p}(\Omega)}, |v|_{W^{1,p}(\Omega)}) \|v\|_{(w,p)},$$

$$(2.5)\qquad \|v\|_{(w,p)}^2 \leq |v|_{W^{1,p}(\Omega)}^p,$$

and for $2 \leq p < \infty$,

$$(2.6)\qquad |v|_{W^{1,p}(\Omega)}^p \leq \|v\|_{(w,p)}^p \leq C(|w|_{W^{1,r}(\Omega)}, |v|_{W^{1,r}(\Omega)}) |v|_{W^{1,s}(\Omega)}^2,$$

where $s \in [2, p]$, $r = s(2-p)/(2-s)$.

In a priori error analysis we almost always take w to be u, the solution of (WP).

The essential relations between the quasi norm and the equation are reflected in the following inequalities, which follow from Lemmas 5.1 and 5.2 in the appendix (see [8] for the proof). Let u be the solution of (WP). For any $v \in W^{1,p}(\Omega)$,

$$(2.7)\qquad a(u, u-v) - a(v, u-v) \geq C\|u-v\|_{(u,p)}^{\rho},$$

and for any $\theta > 0$

$$(2.8)\quad |a(u,w) - a(v,w)| \leq C\theta^{\gamma} \|u-v\|_{(u,p)}^{\rho} + C\theta \|w\|_{(u,p)}^{\rho} \quad \forall v, w \in W^{1,p}(\Omega),$$

where the constant $\gamma > 0$ is defined in Lemma 5.2.

A priori error bounds for the finite element approximation of (WP) can then be easily obtained from (2.7) and (2.8), and summarized in terms of the quasi norm in the following way (see [7] and [8]). For any $v^h \in V_g^h$,

$$(2.9)\qquad \|u - u_h\|_{(u,p)} \leq C\|u - v_h\|_{(u,p)},$$

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where u and u_h are the solutions of (WP) and $(WP)^h$, respectively. This shows that the finite element approximation of the solution of (WP) can be bounded above by the best approximation of the solution by the finite element subspace V_g^h in the quasi norm. As far as we know, there is no such a result in any norm in the literature.

From (2.9) various explicit a priori error bounds can be derived depending on the regularity of the solution u. For instance, if $1 < p \leq 2$ and u is sufficiently regular, then it is shown in [7] using (2.9) that

$$\|u - u_h\|_{W^{1,p}(\Omega)} \leq Ch.$$

It is realized that the $W^{1,p}$ norm is too strong to obtain any optimal a priori error bounds when $p > 2$ (see [7] and [19]). It follows from Proposition 2.1 that the quasi norm is even stronger than the $W^{1,p}$ norm when $p > 2$. In fact, it follows that

$$c\|u - v\|_{(u,p)} \geq |u - v|_{W^{1,p}(\Omega)} + \left(\int_\Omega |\nabla u|^{p-2}|\nabla(u - v)|^2\right)^{1/p}.$$

Therefore, in general, one can expect only that $\|u - \pi_h u\|_{(u,p)}$ is of $h^{2/p}$ as $h \to 0$, no matter how regular the solution u is. This causes few problems in a priori error analysis, as seen from (2.9). However, this quasi norm is not convenient in an a posteriori error analysis when $p > 2$. In this paper we shall use $p/2$ power of the quasi norm to measure the discretization error when $p > 2$, though we still refer it to as a quasi norm.

Let $w \in W^{1,p}(\Omega)$. We define for any $v \in W^{1,p}(\Omega)$

$$|v|_{(w,p)}^2 \equiv \int_\Omega |\nabla v|^2(|\nabla w| + |\nabla v|)^{(p-2)}. \tag{2.10}$$

When $p \leq 2$, $|\cdot|_{(w,p)} = \|\cdot\|_{(w,p)}$. When $p > 2$, $|\cdot|_{(w,p)} = (\|\cdot\|_{(w,p)})^{p/2}$.

It follows from (2.7) and (2.8) that there are constants $c, C > 0$ such that for any $u, v \in W^{1,p}(\Omega)$

$$c(a(u, v - u) - a(v, u - v)) \leq |u - v|_{u,p}^2 \leq C(a(u, u - v) - a(v, u - v)).$$

Thus, it seems that the quasi norm is naturally related to the total energy difference $a(u, u - v) - a(v, u - v)$.

Although, again, we shall usually take w to be u, it is sometimes desirable to replace it with u_h (see [25]) in a posteriori error estimation, where u_h is the finite element approximation of u, in order to make the quasi norm "computable." It follows from the triangle inequality that there are constants $c, C > 0$, independent of h, such that

$$c|u - u_h|_{(u,p)} \leq |u - u_h|_{(u_h,p)} \leq C|u - u_h|_{(u,p)}.$$

We shall simply write $|\cdot|_{(u,p)}$ as $|\cdot|_{(p)}$ when no confusion is possible.

It is clear that (2.7) and (2.8) hold for $|\cdot|_{(u,p)}$ with $\rho = 2$, though the triangle inequality in Proposition 2.1 may not be true for $p > 2$. However, it follows from Lemma 5.4 that there is a $C = C(p) > 0$ such that for any $v_1, v_2, w \in W^{1,p}(\Omega)$,

$$|v_1 + v_2|_{(w,p)} \leq C(|v_1|_{(w,p)} + |v_2|_{(w,p)}). \tag{2.11}$$

It is far from straightforward to extend the quasi-norm techniques used in a priori error analysis to a posteriori error estimation. It seems that many new tools have to

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be established for this purpose. Besides the inequalities in the appendix, we need the following interpolation results for piecewise polynomials of the maximum degree of k. Essentially, these are the quasi-norm version of some well-known results on π_h.

LEMMA 2.1. *Let $K \in T^h$, and let π_h be the standard piecewise linear Lagrange interpolation operator. Let k be a nonnegative integer, and let $v \in H^1(\Omega)$ such that $v|_K$ is a t-degree polynomial with $t \le k$. Then,*

$$(2.12)\quad \int_K (|\nabla u_h| + h_K^{-1}|v - \pi_h v|)^{p-2} h_K^{-2} |v - \pi_h v|^2 \le C \int_K (|\nabla u_h| + |\nabla v|)^{p-2} |\nabla v|^2$$

and

$$(2.13)\quad \int_K (|\nabla u_h| + |\nabla (v - \pi_h v)|)^{p-2} |\nabla (v - \pi_h v)|^2 \le C \int_K (|\nabla u_h| + |\nabla v|)^{p-2} |\nabla v|^2 ,$$

where the constant C depends only on k, the maximum degree of the polynomials, and the regularity constant of T^h.

Proof. By a standard interpolation estimate and the inverse inequality (see [11]), $\forall x \in K$,

$$|v(x) - \pi_h v(x)| \le \|v - \pi_h v\|_{0,\infty,K} \le C h_K |v|_{1,\infty,K} \le C h_K^{1-n} \int_K |\nabla v|,$$

where n is the dimension of Ω. Therefore,

$$\begin{aligned} &\int_K (|\nabla u_h| + h_K^{-1}|v - \pi_h v|)^{p-2} h_K^{-2} |v - \pi_h v|^2 \\ &\quad \le \int_K \left(|\nabla u_h| + \int_K C h_K^{-n} |\nabla v| \right)^{p-2} \left(\int_K C h_K^{-n} |\nabla v| \right)^2 \\ &\quad = meas(K) \left(a + \int_K r \right)^{p-2} \left(\int_K r \right)^2 , \end{aligned}$$

where $a = |\nabla u_h|$ and $r = C h_K^{-n} |\nabla v|$, since the function $x \to s(x) = (a + x)^{p-2} x^2$ is increasing on $[0, \infty)$ for any $a \ge 0$. It is a matter of calculation to show that the function s is further a convex continuous function on $[0, \infty)$. It is also clear that r is continuous on $\bar{K}$. Therefore, it follows from the integral form of Jensen's inequality that

$$meas(K) s \left(\int_K r \right) \le \int_K s(meas(K) r).$$

Then, (2.12) follows from the above two inequalities as there are constants $c, C \ge 0$, independent of h_K and K, such that $c h_K^n \le meas(K) \le C h_K^n$. ∎

It is clear that (2.13) can be proved similarly.

The following result is well known.

LEMMA 2.2 (see [14]). *For all $v \in W^{1,q}(\Omega^h)$, $1 \le q < \infty$,*

$$(2.14)\quad \|v\|_{0,q,\partial K} \le C \left(h_K^{-\frac{1}{q}} \|v\|_{0,q,K} + h_K^{1-\frac{1}{q}} |v|_{1,q,K} \right).$$

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3. A posteriori error estimators. In this section we extend the quasi-norm techniques used in obtaining sharp a priori error bounds for the p-Laplacian in [7] and [19] in order to derive improved a posteriori error estimates for the finite element approximation of the equation. More precisely, we derive a posteriori error estimators with both lower and upper error bounds in the quasi norm $|\cdot|_{(u,p)}$ defined in section 2. For sufficiently regular solutions, these estimators are shown to be equivalent on the discretization error in this quasi norm.

3.1. Upper bounds. First we introduce some notation. Let l be a face of an element $K \in T^h$. If l is on the boundary of Ω^h, then we define the element $K_l = K$. Otherwise, let $l = \bar{K}_l^1 \cap \bar{K}_l^2$, where K_l^1, K_l^2 are the two elements sharing the common face l. Then, we define the element $K_l = K_l^i$ (i=1 or 2) such that

$$|\nabla u_h|_{K_l}^{p-2} = \max_{i=1,2} \left\{ |\nabla u_h|_{K_l^i}^{p-2} \right\}.$$

We will take $K_l = K_l^1$ just for fixing idea if $|\nabla u_h|_{K_l^1}^{p-2} = |\nabla u_h|_{K_l^2}^{p-2}$. The significance of this definition will be seen in the proof of Lemmas 3.1 and 3.2. Let $V_0^h(k)$ denote the space of all continuous piecewise k-order polynomials p_k^h on T^h such that $p_k^h \in H_0^1(\Omega)$. Let $\pi u \in V_0^h(k)$ be a piecewise k-order polynomial approximation of u. Let A_l be the p-normal derivative jump over an interior face l: defined by

$$A_l = \left[|\nabla u_h|^{p-2} \frac{\partial u_h}{\partial n} \right]_l = ((|\nabla u_h|^{p-2} \nabla u_h)_{K_l^1} - (|\nabla u_h|^{p-2} \nabla u_h)_{K_l^2}) \cdot \vec{n},$$

where $l = \bar{K}_l^1 \cap \bar{K}_l^2$ is the common face of two elements K_l^1 and K_l^2, $\vec{n}$ is the unit normal vector of the face l outwards K_l^1. For ease of exposition, first we derive the estimates for the case $g = 0$. Then it is not difficult to extend the results to the case where g may not be zero but is sufficiently regular later on. For later convenience, we shall define the jump A_l to be zero when a face l is on the boundary.

THEOREM 3.1. *Let u and u_h be the solutions of (2.1) and (2.2) with $g = 0$, respectively. Let $p > 1$, and let $p' > 1$ be such that $\frac{1}{p} + \frac{1}{p'} = 1$. Assume that $f \in L^{p'}(\Omega)$. Then, for any fixed integer $k \geq 2$ and $\forall\, \pi u \in V_0^h(k)$*

$$|u - u_h|_{(p)}^2 \leq C(\eta_1^2 + \eta_2^2) + C\epsilon_1,$$

where

$$|u - u_h|_{(p)}^2 = \int_\Omega (|\nabla u| + |\nabla(u - u_h)|)^{p-2} |\nabla(u - u_h)|^2,$$

$$\eta_1^2 = \sum_K \int_K (|\nabla u_h|^{p-1} + h_K |f|)^{p'-2} h_K^2 |f|^2,$$

$$\eta_2^2 = \sum_l \int_{K_l} (|\nabla u_h|^{p-1} + |A_l|)^{p'-2} A_l^2,$$

$$\epsilon_1 = \inf_{\pi u \in V_0^h(k)} |u - \pi u|_{(p)}^2.$$

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Proof. Let $\hat{e} = \pi u - u_h$. First, note that $(a+b)/2 \le a + |a-b| \le 2(a+b)$ $\forall\, a, b \ge 0$. Then, it follows from (2.1), (5.2) (with $\delta = 0$), and (2.2) that

$$
\begin{aligned}
|u-u_h|^2_{(p)} &= \int_\Omega (|\nabla u| + |\nabla(u-u_h)|)^{p-2} |\nabla(u-u_h)|^2 \\
&\le C \int_\Omega (|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h)\nabla(u-u_h) \\
&= C \int_\Omega (|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h)\nabla(u-u_h-\pi_h\hat{e}) \\
&= C \int_\Omega (|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h)\nabla((u-\pi u)+(\hat{e}-\pi_h\hat{e})) \\
&= C \int_\Omega (|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h)\nabla(u-\pi u) \\
&\quad + C\sum_K \int_{\partial K} (|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h)\cdot \vec{n}(\hat{e}-\pi_h\hat{e}) \\
&\quad - C\sum_K \int_K div(|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h)(\hat{e}-\pi_h\hat{e}) \\
&= C\int_\Omega (|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h)\nabla(u-\pi u) \\
&\quad - C\sum_l \int_l \left[|\nabla u_h|^{p-2}\frac{\partial u_h}{\partial n}\right](\hat{e}-\pi_h\hat{e}) + C\sum_K \int_K f(\hat{e}-\pi_h\hat{e}) \\
&= I_1 + I_2 + I_3.
\end{aligned} \tag{3.1}
$$

It follows from Lemmas 5.1 and 5.2 that for any $\theta_1 > 0$

$$
I_1 \le C\theta_1^{-\gamma}|u-u_h|^2_{(p)} + C\theta_1 |u-\pi u|^2_{(p)}, \tag{3.2}
$$

where γ is defined in Lemma 5.2. Then it follows from Lemmas 5.3, 2.1, and 2.2 that for any $\delta_1 > 0$

$$
\begin{aligned}
I_2 &= -C\sum_l \int_l \left[|\nabla u_h|^{p-2}\frac{\partial u_h}{\partial n}\right](\hat{e}-\pi_h\hat{e}) \le C\sum_l \int_{\partial K_l} |A_l|\,|\hat{e}-\pi_h\hat{e}| \\
&\le C\sum_l \int_{K_l} |A_l|\,(h_{K_l}^{-1}|\hat{e}-\pi_h\hat{e}| + |\nabla(\hat{e}-\pi_h\hat{e})|) \\
&\le C\delta_1^{-\beta}\sum_l \int_{K_l} (|\nabla u_h|^{p-1} + |A_l|)^{p'-2} A_l^2 \\
&\quad + C\delta_1 \sum_l \int_{K_l} (|\nabla u_h| + h_K^{-1}|\hat{e}-\pi_h\hat{e}|)^{p-2} h_K^{-2}|\hat{e}-\pi_h\hat{e}|^2 \\
&\quad + C\delta_1 \sum_l \int_{K_l} (|\nabla u_h| + |\nabla(\hat{e}-\pi_h\hat{e})|)^{p-2} |\nabla(\hat{e}-\pi_h\hat{e})|^2 \\
&\le C\delta_1^{-\beta}\eta_2^2 + C\delta_1 \sum_l \int_{K_l} (|\nabla u_h| + |\nabla\hat{e}|)^{p-2}|\nabla\hat{e}|^2,
\end{aligned} \tag{3.3}
$$

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where β is defined in Lemma 5.3. Similarly, from Lemmas 5.3 and 2.1, for any $\delta_2 > 0$

$$
\begin{aligned}
I_3 &= C\sum_K \int_K h_K f h_K^{-1}(\hat{e} - \pi_h \hat{e}) \\
&\le C\delta_2^{-\beta}\sum_K \int_K (|\nabla u_h|^{p-1} + h_K|f|)^{p'-2} h_K^2 |f|^2 \\
&\quad + C\delta_2 \sum_K \int_K (|\nabla u_h| + h_K^{-1}|\hat{e} - \pi_h\hat{e}|)^{p-2} h_K^{-2}|\hat{e} - \pi_h\hat{e}|^2 \\
&\le C\delta_2^{-\beta}\eta_1^2 + C\delta_2 \sum_K \int_K (|\nabla u_h| + |\nabla \hat{e}|)^{p-2}|\nabla \hat{e}|^2.
\end{aligned} \tag{3.4}
$$

It follows from (3.1)–(3.4) that

$$
\begin{aligned}
|u - u_h|_{(p)}^2 &\le C(\eta_1^2 + \eta_2^2) + C|u - \pi u|_{(p)}^2 + C\theta_1^{-\gamma}|u - u_h|_{(p)}^2 \\
&\quad + C(\delta_1 + \delta_2)\sum_K \int_K (|\nabla u_h| + |\nabla \hat{e}|)^{p-2}|\nabla \hat{e}|^2.
\end{aligned} \tag{3.5}
$$

By Lemma 5.5

$$
\begin{aligned}
&\sum_K \int_K (|\nabla u_h| + |\nabla \hat{e}|)^{p-2}|\nabla \hat{e}|^2 \le C\sum_K \int_K (|\nabla u_h| + |\nabla \pi u|)^{p-2}|\nabla(u_h - \pi u)|^2 \\
&\le C\left(\sum_K \int_K (|\nabla u| + |\nabla(u - u_h)|)^{p-2}|\nabla(u - u_h)|^2 \right. \\
&\qquad \left. + \sum_K \int_K (|\nabla u| + |\nabla(u - \pi u)|)^{p-2}|\nabla(u - \pi u)|^2\right) \\
&= C|u - u_h|_{(p)}^2 + C|u - \pi u|_{(p)}^2.
\end{aligned}
$$

Therefore,

$$
|u - u_h|_{(p)}^2 \le C(\eta_1^2 + \eta_2^2) + C\epsilon_1 + C(\theta_1^{-\gamma} + \delta_1 + \delta_2)|u - u_h|_{(p)}^2.
$$

Hence, the theorem follows by letting $\theta_1^{-\gamma} + \delta_1 + \delta_2 = \frac{1}{2C}$. □

One of the main reasons why we do not use a standard averaging interpolator such as the one introduced in [24] is that a quasi-norm version of the H^1 stability result of such averaging interpolators seems rather complicated to establish. For instance, one has to estimate the integrals such as

$$
\int_{K'} (|\nabla u_h|_K + |\nabla(v - \pi_h v)|)^{p-2}|\nabla(v - \pi_h v)|^2,
$$

where K and K' are two elements having a common vertex. Then it seems difficult to bound the $|u - u_h|_{(p)}^2$ with $\eta_1^2 + \eta_2^2$. This seems a technical difficulty in applying quasi-norm techniques in deriving sharper a posteriori error estimates for the degenerate systems. What is needed is an averaging interpolator stable in the quasi norm, which is unfortunately not available yet. Our approach is first to approximate the solution with a (fixed order) piecewise polynomial πu and then to apply the standard Lagrange piecewise linear interpolator π_h to $\pi u - u_h$. Then we use the quasi-norm version π_h

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H^1-stability result (Lemma 2.1) for piecewise polynomials of a fixed order. This simpler approach greatly simplifies the resulting estimates and seems effective at least when the solution is sufficiently regular. However, it does lead to an extra "higher order" term ϵ_1 in the resulting *upper* error bound, though this extra term seems due only to the proof techniques. In fact, it should be, in general, insignificant in numerical implementation; see Remark 3.1 below and the numerical tests in section 4. Furthermore, this term can be removed by modifying the definition of η_2 slightly.

Remark 3.1. It is clear that $\epsilon_1 \le |u - u_h|^2_{(p)}$ as $u_h \in V_0^h(k)$ for any $k \ge 1$. In general, the term ϵ_1 should not affect numerical implementation of an adaptive finite element scheme based on the above estimator. For instance, when u is sufficiently regular, in general, ϵ_1 is of higher order than $|u - u_h|^2_{(p)}$ (or than η_1^2) as $h \to 0$, which is of the optimal order of h^2. For instance, when $n = 2$ and $1 < p \le 2$, $|u - u_h|^2_{(p)}$ is shown in [18] to be of the optimal order h^2 as $h \to 0$ if $u \in W^{1+2/p,p}(\Omega)$ and the integral $\int_\Omega |Hu|^2/|\nabla u|^{2-p}$ is finite, where Hu is the Hessian matrix of u. It is also shown that this regularity condition is indeed achievable. In this case one has

$$|u - \pi u|_{(p)}/h \le C|u - \pi u|^{p/2}_{W^{1,p}(\Omega)}/h = C(|u - \pi u|_{W^{1,p}(\Omega)}/h^{2/p})^{p/2}.$$

Let $\pi u = \pi_2 u$, where π_2 is the standard Lagrange piecewise quadratic interpolator on T^h. Define the continuous linear operator $J_h : W^{1+2/p,p}(\Omega) \to W^{1,p}(\Omega)$ by

$$J_h v = (v - \pi_2 v)/h^{2/p}.$$

It follows from the interpolation estimate of π_2 for fraction order Sobolev spaces that $\{\|J_h\|\}$ is uniformly bounded. On the other hand, $\lim_{h\to 0} J_h v = 0$ for any $v \in C^\infty(\bar\Omega)$ which is dense in $W^{1+2/p,p}(\Omega)$. Therefore, $\forall\, v \in W^{1+2/p,p}(\Omega)$, $\lim_{h\to 0} J_h v = 0$. Thus, we have shown that $\epsilon_1 = o(h^2)$ as $h \to 0$ in this case. In [7], $|u - u_h|^2_{(p)}$ is also shown to be of h^2 when $u \in W^{3,1}(\Omega) \cap C^{2,2/p-1}(\bar\Omega)$. Similarly, one can show that $\epsilon_1 = o(h^2)$ as $h \to 0$ when $u \in W^{3,1}(\Omega)$.

The case $p \ge 2$ can be discussed similarly. For instance, $\epsilon_1 = o(h^2)$ as $h \to 0$, if $u \in W^{1,\infty}(\Omega) \cap W^{2,2}(\Omega)$. In [7] this regularity condition is shown to be a realistic condition to ensure that $|u - u_h|^2_{(p)}$ has the optimal order of h^2.

The numerical experiments in section 4 confirm the above conclusion.

3.2. Lower bounds. For the rest of the paper, we shall assume that $\Omega \subset R^2$, though all the results can be generalized to a three-dimensional domain. To derive a lower bound we need two further lemmas.

LEMMA 3.1. *Let l be an edge shared by two elements $K_l^1, K_l^2 \in T^h$: $l = \bar K_l^1 \cap \bar K_l^2$. Let K_l be defined as before. Then, for any constant A, there exists a function $w_l \in H_0^1(\Omega)$ such that $w_l|_{\Omega\setminus(\bar K_l^1 \cup \bar K_l^2)} = 0$,*

$$\int_l A w_l = \int_{K_l} (|\nabla u_h|^{p-1} + |A|)^{p'-2} A^2, \tag{3.6}$$

$$\|w_l\|_{0,\infty} \le C h_{K_l} (|\nabla u_h|^{p-1}_{K_l} + |A|)^{p'-2} |A|, \tag{3.7}$$

and

$$\int_{K_l^1 \cup K_l^2} (|\nabla u_h| + |\nabla w_l|)^{p-2} |\nabla w_l|^2 \le C \int_{K_l} (|\nabla u_h|^{p-1} + |A|)^{p'-2} A^2. \tag{3.8}$$

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Proof. The lemma clearly holds when $A = 0$ by taking $w_l = 0$. We assume that $A \neq 0$. On any edge l, let λ_1, λ_2 be the base functions of linear triangular elements (barycentric coordinates) for the two vertices on l, and let $w_l = \alpha_l \lambda_1 \lambda_2$. Hence, w_l is a local bubble function for the edge l,(see [4], [27], and [23] for details). Then, it follows that (3.6) and (3.7) hold if one takes

$$\alpha_l = \int_{K_l} (|\nabla u_h|^{p-1} + |A|)^{p'-2} A / \int_l \lambda_1 \lambda_2 \, .$$

Note that

$$\begin{aligned}
& \int_{K_l^1 \cup K_l^2} (|\nabla u_h| + |\nabla w_l|)^{p-2} |\nabla w_l|^2 \\
\leq \; & C \int_{K_l^1 \cup K_l^2} (|\nabla u_h| + |\nabla w_l|)^{p-2} (h_{K_l}^{-1} \|w_l\|_{0,\infty})^2 \\
\leq \; & C \int_{K_l^1 \cup K_l^2} (|\nabla u_h| + |\nabla w_l|)^{p-2} (|\nabla u_h|_{K_l}^{p-1} + |A|)^{2(p'-2)} A^2 \\
= \; & C \int_{K_l^1 \cup K_l^2} \frac{(|\nabla u_h| + |\nabla w_l|)^{p-2}}{(|\nabla u_h|_{K_l}^{p-1} + |A|)^{2-p'}} (|\nabla u_h|_{K_l}^{p-1} + |A|)^{p'-2} A^2 \\
= \; & \frac{C}{|K_l|} \int_{K_l^1 \cup K_l^2} Q^{\frac{p-2}{p-1}} \int_{K_l} (|\nabla u_h|^{p-1} + |A|)^{p'-2} A^2,
\end{aligned}$$

where

$$Q = \frac{(|\nabla u_h| + |\nabla w_l|)^{p-1}}{|\nabla u_h|_{K_l}^{p-1} + |A|}.$$

Then, the lemma will follow if one can show that there is a constant $C > 0$ such that

$$\frac{1}{|K_l|} \int_{K_l^1 \cup K_l^2} Q^{\frac{p-2}{p-1}} \leq C. \tag{3.9}$$

First let us consider the case $1 < p \leq 2$. If $|A| \leq |\nabla u_h|_{K_l}^{p-1}$, then

$$Q^{-1} \leq \max_{i=1,2} \frac{2|\nabla u_h|_{K_l}^{p-1}}{|\nabla u_h|_{K_l^i}^{p-1}} \leq 2.$$

Therefore,

$$\frac{1}{|K_l|} \int_{K_l^1 \cup K_l^2} Q^{\frac{p-2}{p-1}} \leq C,$$

where the constant C depends only on p and the regularity constant of T^h. On the other hand, if $|\nabla u_h|_{K_l}^{p-1} \leq |A|$, then

$$\begin{aligned}
Q^{-1} \; & \leq \; \frac{2|A|}{|\nabla w_l|^{p-1}} = \frac{2|A|}{|\alpha_l \nabla(\lambda_1 \lambda_2)|^{p-1}} \\
& \leq \; \frac{C|A|}{(h_{K_l}(|\nabla u_h|_{K_l}^{p-1} + |A|)^{p'-2}|A| \, |\nabla(\lambda_1 \lambda_2)|)^{p-1}} \\
& \leq \; \frac{C|A|}{(h_{K_l}|A|^{p'-2}|A| \, |\nabla(\lambda_1 \lambda_2)|)^{p-1}} \leq C h_{K_l}^{1-p} |\nabla(\lambda_1 \lambda_2)|^{1-p}.
\end{aligned}$$

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Hence,

$$\int_{K_l^1\cup K_l^2} Q^{\frac{p-2}{p-1}} \le \int_{K_l^1\cup K_l^2} Ch_{K_l}^{p-2}|\nabla(\lambda_1\lambda_2)|^{p-2}$$
$$\le Ch_{K_l}^2 \int_{\hat{K}_l^1\cup\hat{K}_l^2} |\nabla(\hat{\lambda}_1\hat{\lambda}_2)|^{p-2} \le Ch_{K_l}^2,$$

where $\hat{K}_l^i$ is the reference elements of K_l^i, $i = 1, 2$. Then, we have

$$\frac{1}{|K_l|}\int_{K_l^1\cup K_l^2} Q^{\frac{p-2}{p-1}} \le \frac{Ch_{K_l}^2}{|K_l|} \le C.$$

Hence, (3.9) follows for $1 < p \le 2$. Now consider the case $p \ge 2$. For $i = 1$ or 2, if $\|\nabla w_l\|_{0,\infty,K_l^i} \le |\nabla u_h|_{K_l^i}$, then

$$Q \le \frac{(2|\nabla u_h|_{K_l^i})^{p-1}}{|\nabla u_h|_{K_l}^{p-1}} \le (2)^{p-1}.$$

If $|\nabla u_h|_{K_l^i} \le \|\nabla w_l\|_{0,\infty,K_l^i}$, then

$$Q \le \frac{(2\|\nabla w_l\|_{0,\infty,K_l^i})^{p-1}}{|A|} \le C\frac{((|\nabla u_h|_{K_l}^{p-1} + |A|)^{p'-2}|A|)^{p-1}}{|A|} \le \frac{C(|A|^{p'-2}|A|)^{p-1}}{|A|} \le C.$$

Therefore, again

$$\frac{1}{|K_l|}\int_{K_l^1\cup K_l^2} Q^{\frac{p-2}{p-1}} \le C\frac{|K_l^1\cup K_l^2|}{|K_l|} \le C.$$

This proves (3.9) for $p \ge 2$. Hence, the lemma follows. □

Similarly, we can prove the following lemma.

LEMMA 3.2. *For any element $K \in T^h$ and any constant f, there exists a polynomial w_K on K such that $w_K|_{\partial K} = 0$,*

$$\int_K fw_K = \int_K (|\nabla u_h|^{p-1} + h_K|f|)^{p'-2}h_K^2|f|^2,$$

$$\|w_K\|_{0,\infty} \le C(|\nabla u_h|^{p-1} + h_K|f|)^{p'-2}h_K^2|f|,$$

and

$$\int_K (|\nabla u_h| + |\nabla w_K|)^{p-2}|\nabla w_K|^2 \le C\int_K (|\nabla u_h|^{p-1} + h_K|f|)^{p'-2}h_K^2|f|^2.$$

Proof. Let w_K be the following function: $w_K = \alpha_K\lambda_1\lambda_2\lambda_3$ (see [4], [27], and [23]), and let

$$\alpha_K = \frac{\int_K (|\nabla u_h| + h_K|f|)^{p'-2}h_K^2|f|^2}{\int_K f\lambda_1\lambda_2\lambda_3}.$$

Then, one can check as in Lemma 3.1 that all the conclusions in the lemma hold. □

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We are now in the position of deriving a lower bound for the estimate given in Theorem 3.1.

THEOREM 3.2. *Let u and u_h be the solutions of (2.1) and (2.2), respectively. Let $p > 1$ and $p' > 1$ be such that $\frac{1}{p} + \frac{1}{p'} = 1$. Assume that $f \in L^{p'}(\Omega)$. Then,*

$$\eta_1^2 + \eta_2^2 \le C|u - u_h|_{(p)}^2 + C\epsilon_2,$$

where

$$\epsilon_2 = \sum_K \int_K (|\nabla u_h|^{p-1} + h_K|f - \bar{f}|)^{p'-2} h_K^2 |f - \bar{f}|^2,$$

where $\bar{f}|_K = \int_K f/|K|$.

Proof. It follows from Lemmas 5.1, 5.2, and 3.1 that for any $\theta_1, \theta_2 > 0$

$$\begin{aligned}
\eta_2^2 &= \sum_l \int_{K_l} (|\nabla u_h|^{p-1} + |A_l|)^{p'-2} A_l^2 = \sum_l \int_l A_l w_l \\
&= \sum_l \int_l \left[|\nabla u_h|^{p-2} \frac{\partial u_h}{\partial n} - |\nabla u|^{p-2} \frac{\partial u}{\partial n} \right]_l w_l \\
&= \sum_l \int_{\partial K_l^1 \cup \partial K_l^2} \left(|\nabla u_h|^{p-2} \frac{\partial u_h}{\partial n} - |\nabla u|^{p-2} \frac{\partial u}{\partial n} \right) w_l \\
&= \sum_l \left(\int_{K_l^1 \cup K_l^2} (|\nabla u_h|^{p-2} \nabla u_h - |\nabla u|^{p-2} \nabla u) \nabla w_l + \int_{K_l^1 \cup K_l^2} f w_l \right) \\
&\le C \sum_l \int_{K_l^1 \cup K_l^2} (|\nabla u_h| + |\nabla (u - u_h)|)^{p-2} |\nabla (u_h - u)| \, |\nabla w_l| \\
&\quad + C \sum_l \int_{K_l^1 \cup K_l^2} |f| h_{K_l} (|\nabla u_h|_{K_l}^{p-1} + |A_l|)^{p'-2} |A_l| \\
&\le C\theta_1^{-\gamma} \sum_l \int_{K_l^1 \cup K_l^2} (|\nabla u_h| + |\nabla (u_h - u)|)^{p-2} |\nabla (u_h - u)|^2 \\
&\quad + C\theta_1 \sum_l \int_{K_l^1 \cup K_l^2} (|\nabla u_h| + |\nabla w_l|)^{p-2} |\nabla w_l|^2 \\
&\quad + C\theta_2^{-\gamma} \sum_l \int_{K_l^1 \cup K_l^2} (|\nabla u_h|_{K_l}^{p-1} + |A_l|)^{p'-2} A_l^2 \\
&\quad + C\theta_2 \sum_l \int_{K_l^1 \cup K_l^2} (|\nabla u_h|_{K_l}^{p-1} + h_{K_l} |f|)^{p'-2} h_{K_l}^2 |f|^2 \\
&\le C\theta_1^{-\gamma} |u - u_h|_{(p)}^2 + C\theta_2 \eta_1^2 + C(\theta_1 + \theta_2^{-\gamma}) \eta_2^2,
\end{aligned}$$

where γ is defined in Lemma 5.2.

Let $\theta_1 = \frac{1}{4C}$, $\theta_2 = (4C)^{\frac{1}{\gamma}}$. Then, we have

$$\eta_2^2 \le C|u - u_h|_{(p)}^2 + C\eta_1^2. \tag{3.10}$$

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From Lemma 5.4,

$$
\begin{aligned}
\eta_1^2 &= \sum_K \int_K (|\nabla u_h|^{p-1} + h_K|f|)^{p'-2} h_K^2 |f|^2 \\
&\le C \sum_K \int_K (|\nabla u_h|^{p-1} + h_K|\bar{f}|)^{p'-2} h_K^2 |\bar{f}|^2 \\
&\quad + C \sum_K \int_K (|\nabla u_h|^{p-1} + h_K|f-\bar{f}|)^{p'-2} h_K^2 |f-\bar{f}|^2 \\
&= I + C\epsilon_2,
\end{aligned} \tag{3.11}
$$

where we have simply written $\bar{f}|_K$ as $\bar{f}$. It follows from Lemma 3.2 that

$$
\begin{aligned}
I &= C \sum_K \int_K (|\nabla u_h|^{p-1} + h_K|\bar{f}|)^{p'-2} h_K^2 |\bar{f}|^2 \\
&= \sum_K \int_K \bar{f} w_K = \sum_K \int_K f w_K + \sum_K \int_K (\bar{f} - f) w_K \\
&= I_1 + I_2.
\end{aligned} \tag{3.12}
$$

It follows from Lemmas 5.1, 5.2, 5.4, and 3.2 that for any $\theta > 0$

$$
\begin{aligned}
I_1 &= \sum_K \int_K f w_K = -\sum_K \int_K div(|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h) w_K \\
&= \sum_K \int_K (|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h) \nabla w_K \\
&\le C \sum_K \int_K (|\nabla u_h| + |\nabla (u-u_h)|)^{p-2} |\nabla (u-u_h)| \, |\nabla w_K| \\
&\le C\theta_1^{-\gamma} \sum_K \int_K (|\nabla u_h| + |\nabla (u-u_h)|)^{p-2} |\nabla (u-u_h)|^2 \\
&\quad + C\theta_1 \sum_K \int_K (|\nabla u_h| + |\nabla w_K|)^{p-2} |\nabla w_K|^2 \\
&\le C\theta_1^{-\gamma} |u-u_h|_{(p)}^2 + C\theta_1 \sum_K \int_K (|\nabla u_h|^{p-1} + h_K|\bar{f}|)^{p'-2} h_K^2 |\bar{f}|^2 \\
&= C\theta_1^{-\gamma} |u-u_h|_{(p)}^2 + C\theta_1 I.
\end{aligned} \tag{3.13}
$$

It follows from Lemmas 3.2 and 5.2 that for any $\delta > 0$

$$
\begin{aligned}
I_2 &= \sum_K \int_K (\bar{f} - f) w_K \le \sum_K \int_K |f - \bar{f}| \, \|w_K\|_{0,\infty} \\
&\le C \sum_K \int_K |f-\bar{f}| (|\nabla u_h|^{p-1} + h_K|\bar{f}|)^{p'-2} h_K^2 |\bar{f}| \\
&\le C\theta_2^{-\gamma} \sum_K \int_K (|\nabla u_h|^{p-1} + h_K|\bar{f}|)^{p'-2} h_K^2 |\bar{f}|^2 \\
&\quad + C\theta_2 \sum_K \int_K (|\nabla u_h|^{p-1} + h_K|f-\bar{f}|)^{p'-2} h_K^2 |f-\bar{f}|^2 \\
&\le C\theta_2^{-\gamma} I + C\theta_2 \epsilon_2.
\end{aligned} \tag{3.14}
$$

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From (3.12)–(3.14),

$$I \le C\theta_1^{-\gamma}|u-u_h|_{(p)}^2 + C(\theta_1+\theta_2^{-\gamma})I + C\theta_2\epsilon_2.$$

Let $\theta_1+\theta_2^{-\gamma} = \frac{1}{2C}$. Then

$$I \le C|u-u_h|_{(p)}^2 + C\epsilon_2. \tag{3.15}$$

It follows from (3.11) and (3.15) that

$$\eta_1^2 \le C|u-u_h|_{(p)}^2 + C\epsilon_2. \tag{3.16}$$

Therefore, Theorem 3.2 follows from (3.10) and (3.16). □

Remark 3.2. If f is piecewise constant, then $\epsilon_2 = 0$. In general, the term ϵ_2 is again of higher order than $|u-u_h|_{(p)}$ as $h \to 0$, provided f is smooth enough. For instance, $\epsilon_2 = o(h^2)$ if $f \in W^{\beta,p'}(\Omega)$, provided $\beta > 0$ when $1 < p \le 2$, or $\beta > 1 - \frac{2}{p}$ if $p \ge 2$. Note that ϵ_1 does not appear in the lower bound.

Therefore, we conclude that the estimator $\eta_1+\eta_2$ is equivalent to the discretization error in the quasi norm at least when the solution is sufficiently regular. When $|\nabla u| \neq 0$, for instance, we can infer that estimator $\eta_1+\eta_2$ is equivalent to the discretization error in the conventional $W^{1,p}(\Omega)$ norm.

4. Some extensions and numerical experiments. It is not very difficult to extend the estimates in Theorem 3.1 to the case where g may not be zero but is sufficiently smooth. The following is just an example. Let $V_g^h(k)$ denote the space of all continuous piecewise k-order polynomials p_k^h on T^h such that $p_k^h(P_i) = g(P_i)$, where $\{P_i\}(i = 1 \dots J)$ is the vertex on the boundary $\partial\Omega$ associated with the partitioning T^h. Then we have the following proposition.

PROPOSITION 4.1. *Let u and u_h be the solutions of* (2.1) *and* (2.2) *with $g \in W^{2,\infty}(\partial\Omega)$, respectively. Let $p > 1$ and $p' > 1$ be such that $\frac{1}{p}+\frac{1}{p'} = 1$. Assume that $f \in L^{p'}(\Omega)$. Then, for any fixed integer $k \ge 2$ and $\forall\, \pi u \in V_g^h(k)$*

$$|u-u_h|_{(p)}^2 \le C(\eta_1^2+\eta_2^2) + C\epsilon_1 + C\hat{\epsilon}_1,$$

where

$$|u-u_h|_{(p)}^2 = \int_\Omega (|\nabla u| + |\nabla(u-u_h)|)^{p-2}|\nabla(u-u_h)|^2,$$

$$\eta_1^2 = \sum_K \int_K (|\nabla u_h|^{p-1} + h_K|f|)^{p'-2}h_K^2|f|^2,$$

$$\eta_2^2 = \sum_l \int_{K_l} (|\nabla u_h|^{p-1} + |A_l|)^{p'-2}A_l^2,$$

$$\epsilon_1 = \inf_{p_k^k \in V_g^h(k)} |u-\pi u|_{(p)}^2$$

and

$$\hat{\epsilon}_1 = h^{p+1}|g|_{2,\infty,\partial\Omega} + h^{2.5}\|f\|_{0,\Omega} \quad \textit{when } 1 < p \le 2,$$

$$\hat{\epsilon}_1 = h^{2+2/p}|u|_{1,p,\Omega}^{p-2}|g|_{2,\infty,\partial\Omega} + h^{2.5}\|f\|_{0,\Omega} \quad \textit{when } p > 2.$$

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Proof. Let

$$D = \{\cup K, l \subset \partial K, l \cap \partial\Omega \neq \emptyset\}.$$

We can construct a function $u^* \in H^1(\Omega)$ (see [19] for the detail construct method) such that $u^* = u$ in $\Omega \setminus D$ and $u^* = g_I$ on $\partial\Omega$, and

$$\|u - u^*\|_{0,\infty,\Omega} \leq Ch^2|g|_{2,\infty,\partial\Omega}, \quad |u - u^*|_{1,\infty,\Omega} \leq Ch|g|_{2,\infty,\partial\Omega}.$$

Letting $e^* = u^* - u_h$, $e = u - u_h$, we have

$$\begin{aligned}
|u - u_h|^2_{(p)} &\leq (|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h, \nabla e) \\
&= (|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h, \nabla(e - e_I)) \\
&= (|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h, \nabla(e^* - e_I)) \\
&\quad +(|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h, \nabla(u - u^*)).
\end{aligned}$$

Note that $e^* \in H^1_0(\Omega)$. In the same way as Theorem 3.1, we have

$$\begin{aligned}
&(|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h, \nabla(e^* - e_I)) \\
&= \sum_K \int_K f(e^* - e_I) - \sum_{l\cap\partial\Omega=\emptyset} \int_l \left[|\nabla u_h|^{p-2}\frac{\partial u_h}{\partial n}\right](e^* - e_I)dl \\
&= \sum_K \int_K f(e - e_I) - \sum_{l\cap\partial\Omega=\emptyset} \int_l \left[|\nabla u_h|^{p-2}\frac{\partial u_h}{\partial n}\right](e - e_I)dl + \sum_K \int_K f(u^* - u)
\end{aligned}$$

and

$$\sum_K \int_K f(e - e_I) - \sum_{l\cap\partial\Omega=\emptyset} \int_l \left[|\nabla u_h|^{p-2}\frac{\partial u_h}{\partial n}\right](e - e_I)dl \leq C(\eta_1^2 + \eta_2^2 + \epsilon_1) + \delta|e|^2_{(p)}.$$

Moreover,

$$\begin{aligned}
\sum_K \int_K f(u^* - u) &\leq \|f\|_{0,D}\|u - u^*\|_{0,D} \leq \|f\|_{0,\Omega}\|u - u^*\|_{0,\infty,D}(meas(D))^{\frac{1}{2}} \\
&\leq Ch^{2.5}|g|_{2,\infty.\partial\Omega}\|f\|_{0,\Omega}, (|\nabla u|^{p-2}\nabla u - |\nabla u_h|^{p-2}\nabla u_h, \nabla(u - u^*)) \\
&\leq C|u - u^*|^2_{(p)} + \delta|u - u_h|^2_{(p)}.
\end{aligned}$$

Thus when $1 < p \leq 2$,

$$|u - u^*|^2_{(p)} \leq |u - u^*|^p_{1,p,\Omega} \leq |u - u^*|^p_{1,\infty,\Omega} meas(D) \leq Ch^{1+p}|g|^p_{2,\infty,\Omega},$$

and when $p > 2$,

$$\begin{aligned}
|u - u^*|^2_{(p)} &\leq C|u|^{p-2}_{1,p,\Omega}|u - u^*|^2_{1,p,\Omega} \leq C|u|^{p-2}_{1,p,\Omega}|u - u^*|^2_{1,\infty,\Omega}(meas(D))^{\frac{2}{p}} \\
&\leq Ch^{2+\frac{2}{q}}|g|^2_{2,\infty,\Omega}|u|^{p-2}_{1,p,\Omega}.
\end{aligned}$$

Then, Proposition 4.1 follows. □

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The case $f \notin L^{p'}(\Omega)$ $(1 < p < 2)$. Suppose that $1 < p < 2$. Let us consider the case where $f \in L^2(\Omega)$ but $f \notin L^{p'}(\Omega)$, which is frequently met in applications. For ease of exposition, we assume $\Omega \subset R^2$ and that there are constants $c, C > 0$ such that $(0 < \sigma < 1)$

$$cr^{-\sigma} \le |f| \le Cr^{-\sigma} \quad \text{in } \Omega,$$

where $r = (x^2 + y^2)^{1/2}$. We also assume that $(0,0) \in \Omega$. Therefore, $f \notin L^{p'}(\Omega)$ if $\sigma \ge 2/p'$. It is clear that η_1 or $\hat{\eta}_1$ is not directly applicable for such a case. In order to apply these a posteriori error estimates in numerical implementation, one may consider solving the following truncated approximation: Find $u_h^* \in V_g^h$ such that

$$\int_{\Omega^h} |\nabla u_h^*|^{p-2} \nabla u_h^* \cdot \nabla v_h = \int_{\Omega^h} f^* v_h \quad \forall v_h \in V_0^h, \tag{4.1}$$

where f^* is a truncation of f. As a matter of fact, the Gaussian quadrature scheme may also be considered as a truncation approximation of the problem. Let us examine one of the simplest truncations of f:

$$f^* = \begin{cases} f & \text{in} \quad \Omega \setminus K^*, \\ h^{-\sigma} & \text{in} \quad K^*, \end{cases}$$

where $K^* \in T^h$ and the singular point $(0,0) \in \bar{K}^*$. Then it is desirable to derive a posteriori error estimates for $|u - u_h^*|_{1,p,\Omega}$ to construct an adaptive scheme. One of the possible approaches is to utilize Theorem 3.1. To this end, let u^* be the solution of the following truncated p-Laplacian: $u^* \in W_g^{1,p}(\Omega)$.

$$a(u^*, v) = (f^*, v) \quad \forall v \in W_0^{1,p}(\Omega). \tag{4.2}$$

Then we have the following proposition.

PROPOSITION 4.2. *Let f, f^* be defined as above. Let u be the solution of (WP) and let u^*, u_h^* be defined as above. Then*

$$|u_h^* - u|_{1,p,\Omega} \le |u_h^* - u^*|_{1,p,\Omega} + o(h) \, .$$

Proof. It is clear that

$$|u_h^* - u|_{1,p,\Omega} \le |u_h^* - u^*|_{1,p,\Omega} + |u^* - u|_{1,p,\Omega} \, . \tag{4.3}$$

First let us estimate $|u^* - u|_{1,p,\Omega}$. Let $s = \frac{2p}{3p-2} < 2$ and $s' = \frac{2p}{2-p}$. Then, it follows that $W^{1,p}(\Omega) \hookrightarrow L^{s'}(\Omega)$. Therefore, it follows from (2.1), (4.2), and Lemma 5.1 that

$$\begin{aligned} \|u - u^*\|_{1,p,\Omega}^p &\le C \int_\Omega (|\nabla u|^{p-2} \nabla u - |\nabla u^*|^{p-2} \nabla u^*) \nabla (u - u^*) \\ &= C \int_\Omega (f - f^*)(u - u^*) \le C \|f - f^*\|_{0,s,\Omega} \|u - u^*\|_{0,s',\Omega} \le C \|f \\ &\quad - f^*\|_{0,s,\Omega} \|u - u^*\|_{1,p,\Omega}. \end{aligned}$$

Hence,

$$\|u - u^*\|_{1,p,\Omega}^{p-1} \le C \|f - f^*\|_{0,s,\Omega} \le C \|f\|_{0,s,K^*} + \|f^*\|_{0,s,K^*}.$$

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It is a simple matter to check that

$$\|f\|^s_{0,s,K^*} = \int_{K^*} |f|^s \le C \int_0^h r^{-\sigma s} r dr = Ch^{2-\sigma s}$$

and

$$\|f^*\|^s_{0,s,K^*} = meas(K^*) h^{-\sigma s} = Ch^{2-\sigma s}.$$

Therefore,

$$\|u - u^*\|_{1,p,\Omega} \le Ch^{\rho}, \tag{4.4}$$

where

$$\rho = \frac{2-\sigma s}{s(p-1)} = 1 + \frac{1-\sigma}{p-1} + \frac{2-p}{p} > 1 .$$

Then the proposition follows. □

From Proposition 4.2 one may simply use a posteriori error estimates of $u^* - u_h^*$ as error indicators in an adaptive finite element approximation of u. It also follows from Proposition 4.2 and the existing optimal a priori error bound that we should have $\|u_h^* - u\|_{1,p,\Omega} \le Ch$, provided the solution u^* is sufficiently regular; see Example 4.1.

First let us apply the existing estimate $\hat{\eta} = \hat{\eta}_1 + \hat{\eta}_2$. It is easy to see that

$$\hat{\eta}_1^{p'} \ge Ch_{K^*}^{p'} \int_{K^*} h_{K^*}^{-\sigma p'} = Ch^{p'+2-\sigma p'}.$$

Therefore, if $\sigma > \frac{2}{p'}$, that is, if $f \notin L^{p'}(\Omega)$, then

$$\hat{\eta}_1 \ge Ch^{1-\sigma+\frac{2}{p'}} >> Ch, \tag{4.5}$$

no matter how regular the solutions u and u^* are. This indicates that the a posteriori estimate $\hat{\eta}$ and the real error $u - u_h^*$ may even have different orders as $h \to 0$ from the discussion above. This is also confirmed by one of our numerical experiments in section 4, where $p = 1.5, \sigma = 0.97$. Thus, $\hat{\eta}_1 \ge Ch^{1-\sigma+2/p'} = Ch^{0.697}$. One of our numerical tests indicates that $\hat{\eta}$ is of the order $h^{0.69}$ though $\|u_h^* - u\|_{1,p,\Omega} = O(h)$.

The case for η is different, thanks to the structure of the dominated part of η_1. It will reflect the regularity of the solutions. If, for instance, $|\nabla u_h^*|$ is uniformly bounded above, then

$$\begin{aligned} \eta_1^2 &= \sum_K \int_K (|\nabla u_h^*|^{p-1} + h_K |f^*|)^{p'-2} h_K^2 |f^*|^2 \\ &\le C \sum_K \int_K |\nabla u_h^*|^{2-p} h_K^2 |f^*|^2 + C \sum_K \int_K h_K^{p'} |f^*|^{p'} \\ &\le Ch^2 \textstyle\int_\Omega |f^*|^2 + Ch_{K^*}^{p'} \int_{K^*} h_{K^*}^{-\sigma p'} + Ch^{p'} \int_h^L r^{-\sigma p'} r dr \\ &\le Ch^2 + Ch^{2+p'(1-\sigma)} \le Ch^2 . \end{aligned} \tag{4.6}$$

This is also confirmed in one of our numerical experiments. This may indicate that η may be much sharper that $\hat{\eta}$ in some cases.

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Remark 4.1. It is desirable to compare the existing estimates in the literature with those in Theorem 3.1. Let $1 < p \leq 2$. From [6], [23], and [25]

$$|u - u_h|_{1,p,\Omega} \leq C(\hat{\eta}_1 + \hat{\eta}_2),$$

where

$$\hat{\eta}_1 = \left(\sum_K \int_K (h_K |f|)^{p'} \right)^{1/p'} = \left(\sum_K \|h_K f\|_{0,p',K}^{p'} \right)^{1/p'},$$

$$\hat{\eta}_2 = \left(\sum_l \int_l h_l |A_l|^{p'} \right)^{1/p'}.$$

It follows from Proposition 2.1 that

$$|u - u_h|_{1,p,\Omega} \leq C |u - u_h|_{(p)}.$$

Using a Holder inequality we have

$$\begin{aligned} \eta_1 &= \left(\sum_K \int_K (|\nabla u_h|^{p-1} + h_K |f|)^{p'-2} h_K^2 |f|^2 \right)^{1/2} \\ &\leq C \left(\sum_K |u_h|_{1,p,K}^{2-p} \|h_K f\|_{0,p',K}^2 \right)^{1/2} + C \left(\sum_K \|h_K f\|_{0,p',K}^{p'} \right)^{1/2}. \end{aligned}$$

The term $(\sum_K \|h_K f\|_{0,p',K}^{p'})^{1/2}$ is clearly of higher order than $\hat{\eta}_1$ as $h \to 0$. Also by a Holder inequality

$$\left(\sum_K |u_h|_{1,p,K}^{2-p} \|h_K f\|_{0,p',K}^2 \right)^{1/2} \leq C(|u_h|_{W^{1,p}(\Omega)}^{2-p}) \hat{\eta}_1.$$

Therefore, one has

$$\eta_1 \leq C \hat{\eta}_1.$$

Similarly, one can show

$$\eta_2 \leq C \hat{\eta}_2.$$

Therefore, $\eta_1 + \eta_2$ is certainly not worse than $\hat{\eta}_1 + \hat{\eta}_2$. To see why η_1 may be sharper than $\hat{\eta}_1$ in some cases, let us further assume that $|\nabla u_h|^{1-p/2} \leq C$ and $h = h_K$ $\forall K \in T^h$. Then

$$\hat{\eta}_1 = h \|f\|_{L^{p'}(\Omega)}, \quad \eta_1 \leq Ch \|f\|_{L^2(\Omega)} + \hat{\eta}_1^{p'/2}.$$

Note that $p'/2 > 1$ and $\hat{\eta}_1 \to 0$. Also note that if f has singular points but is in $L^2(\Omega)$, then

$$\|f\|_{L^{p'}(\Omega)} \to \max_{\Omega} |f| = \infty, \quad p \to 1,$$

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though $\|f\|_{L^2(\Omega)}$ remains bounded. For instance, if $f = 1/r^\alpha$ $(0 < \alpha < 1)$ with $r = (x^2 + y^2)^{1/2}$, then $f \notin L^{p'}(\Omega)$ when $2/p' = 2(p-1)/p < \alpha$, though we always have $f \in L^2(\Omega)$. Therefore, one may anticipate that $\hat{\eta}_1$ could be much larger than η_1 when p closes to these values. This is apparently confirmed by our numerical tests. One can have similar discussions for η_2 and $\hat{\eta}_2$. It is also useful to note that $\eta_1 \approx \eta_1' + (\sum_K \|h_k f\|_{0,p',K}^{p'})^{1/2}$, where $\eta_1' = (\sum_K \||\nabla u_h|^{(2-p)/2} h_k f\|_{0,p',K}^2)^{1/2}$. Therefore, η_1' is the dominated part. Thus, we can use just $\eta_1' + \eta_2$ as an error indicator which is finite as long as $f \in L^2(\Omega)$. This character is very useful in dealing with the case where $f \notin L^{p'}(\Omega)$. This will be discussed later.

When $p \geq 2$, it is more convenient to use $|\cdot|_{1,p,\Omega}^{p/2}$ for the comparison following the discussion in section 2. It follows from [6], [23], and [25] that

$$|u - u_h|_{1,p,\Omega}^{p/2} \leq C(\hat{\eta}_1^{p'/2} + \hat{\eta}_2^{p'/2}),$$

where

$$\hat{\eta}_1 = \left(\sum_K \int_K (h_K |f|)^{p'} \right)^{1/p'},$$

$$\hat{\eta}_2 = \left(\sum_l \int_l h_l |A_l|^{p'} \right)^{1/p'}.$$

It follows from Proposition 2.1 that

$$|u - u_h|_{1,p,\Omega}^{p/2} \leq C |u - u_h|_{(p)}.$$

It is clear that

$$\eta_1^2 = \sum_K \int_K (|\nabla u_h|^{p-1} + h_K |f|)^{p'-2} h_K^2 |f|^2 \leq \sum_K \int_K (h_K |f|)^{p'} = \hat{\eta}_1^{p'}.$$

Similarly, we can show that

$$\eta_2^2 \leq \hat{\eta}_2^{p'}.$$

Therefore, similar conclusions hold for the case $p \geq 2$ as well.

Now we report some numerical experiments. We consider here the model problem (2.1) with Dirichlet boundary condition and its finite element approximation (2.2). For computational ease we take Ω to be the square $[0,1] \times [0,1]$. In Examples 4.1–4.3, this is partitioned into uniform right-angled triangles (see [7] for the details) the lengths of whose edges are 1/N, 1/N, and $\sqrt{2}$/N, respectively. We compute our approximation by solving the equivalent minimization problem using a Polak–Ribiére conjugate gradient method. A numerical integration on barycenter (Gauss point of order one) is used for computing the integration.

For Examples 4.1–4.3, we consider the solution of the radially symmetric problem extended to the unite square:

$$u \equiv U(r) \equiv \frac{p-1}{p-\sigma} \left(\frac{1}{2-\sigma} \right)^{\frac{1}{p-1}} (1 - r^{\frac{p-\sigma}{p-1}}),$$

$$f \equiv F(r) \equiv r^{-\sigma}.$$

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TABLE 4.1

N	η	Ratio	$\hat{\eta}$	Ratio	$\|u-u_h\|_{1,p}$	Ratio
4	0.96794125		0.81623822		0.11657411	
8	0.50490126	1.92	0.51211521	1.59	0.06148134	1.90
16	0.26101057	1.94	0.31883059	1.60	0.03180781	1.93
32	0.13414269	1.95	0.19763436	1.61	0.01631292	1.95
64	0.06866920	1.96	0.12225527	1.62	0.00823524	1.98
Order	$O(h^{0.96})$		$O(h^{0.69})$		$O(h^{0.96})$	

TABLE 4.2

N	η	$\hat{\eta}$	$\|u-u_h\|_{1,p}$
4	0.17094209	0.15898636	0.03997645
8	0.08486786	0.08121314	0.02001907
16	0.04230150	0.04110072	0.01005380
32	0.02107880	0.02061937	0.00500672
64	0.01053031	0.01034060	0.00251834
Ratio	2.01	1.98	2.00
Order	$O(h)$	$O(h)$	$O(h)$

TABLE 4.3

N	η	$\hat{\eta}$	$\|u-u_h\|_{1,1}$	$\|u-u_h\|_{1,p}$
4	0.22253180	0.53298512	0.04085823	0.04164706
8	0.11805347	0.32738803	0.02044073	0.02086806
16	0.06080487	0.19765978	0.01022163	0.01044008
32	0.03085870	0.11840028	0.00511138	0.00522187
64	0.01554419	0.07065477	0.00255768	0.00261537
Ratio	1.96	1.66	2.00	2.00
Order	$O(h^{0.97})$	$O(h^{0.73})$	$O(h)$	$O(h)$

In Examples 4.1–4.3, we take $\sigma = 0.97$. We take either $r = \sqrt{x^2+y^2}$ (in Example 4.1) or $r = \sqrt{(x+1)^2+(y+1)^2}$ (in Examples 4.2 and 4.3). It is a simple matter to check that $u \in W^{1,\infty}(\Omega) \cap W^{2,s}(\Omega) \cap W^{3,1}(\Omega)$ with $s > 2$ when $r = \sqrt{x^2+y^2}$ and that $u \in C^\infty(\bar{\Omega})$ when $r = \sqrt{(x+1)^2+(y+1)^2}$.

In Tables 4.1–4.3, we exhibit the numerical results for Examples 4.1–4.3, where η denotes our posteriori error estimate, $\hat{\eta} = \hat{\eta}_1 + \hat{\eta}_2$ when $1 < p < 2$, and $\hat{\eta} = \hat{\eta}_1^{p'/2} + \hat{\eta}_2^{p'/2}$ when $p > 2$; see the discussions in Remark 4.1. In these tables we also calculate the ratio $S_{(1/2^m)}/S_{(1/2^{m+1})}$ $(m = 2, \ldots, 6)$, where $S = \eta$, or $S = \hat{\eta}$, or $S = |u-u_h|_{1,p,\Omega}$. Then we take the average of these ratios in order to estimate the order of the respective quantity when $h \to 0$.

Example 4.1. In this example, $p = 1.5$ and $\sigma = 0.97$. It follows that $f \notin L^{p'}(\Omega)$. Then we use a truncation procedure as suggested before. It is found that η and $|u-u_h^*|_{1,p,\Omega}$ exhibit the order of h as $h \to 0$. This is expected from the existing a priori error bounds and Theorems 3.1–3.2. It is also found that $\hat{\eta}$ exhibits the order of $h^{0.69}$ as $h \to 0$, as predicted. It seems that η is much sharper than $\hat{\eta}$ in this case.

Example 4.2. In this example, again $p = 1.5$ and $\sigma = 0.97$, though $r = \sqrt{(x+1)^2+(y+1)^2}$ so that the solution is infinitely smooth. It is found that all three quantities exhibit the order of h when $h \to 0$. Also η and $\hat{\eta}$ are more or less the same.

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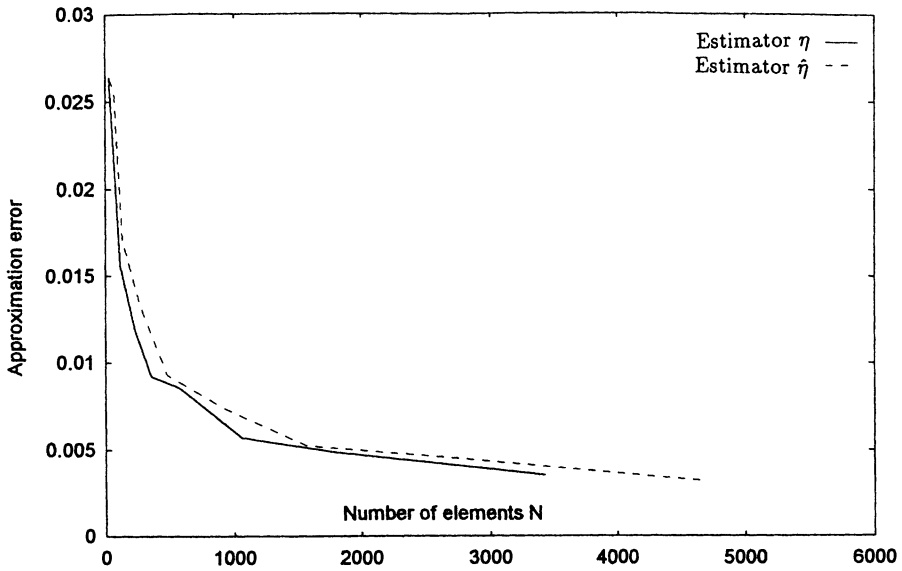


FIG. 1. *Error plots for the solution in Example* 4.1 ($\sigma = 0.8$) *and the meshes generated by the error estimates* η *and* $\hat{\eta}$.

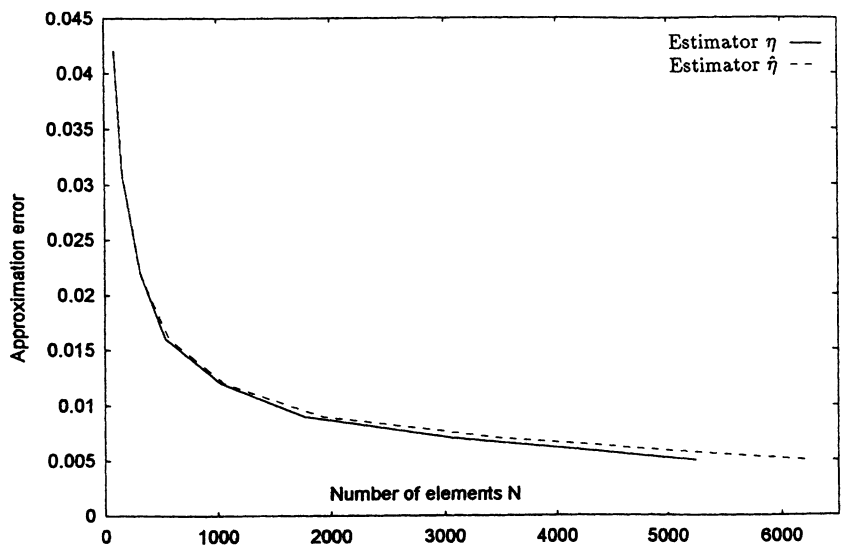


FIG. 2. *Error plots for the solution in Example* 4.1 ($\sigma = 0.97$) *and the meshes generated by the error estimates* η *and* $\hat{\eta}$.

Example 4.3. In this example, $p = 3$ and $\sigma = 0.97$. As in Example 4.2, $r = \sqrt{(x+1)^2 + (y+1)^2}$ so that the solution is sufficiently smooth again. Although η and $|u - u_h|_{1,p,\Omega}$ all exhibit the order of h as $h \to 0$, $\hat{\eta}$ only exhibits the order of $h^{0.73}$ as $h \to 0$. This seems to indicate that η is much sharper in this case.

In the following we report some numerical tests on the performance of mesh

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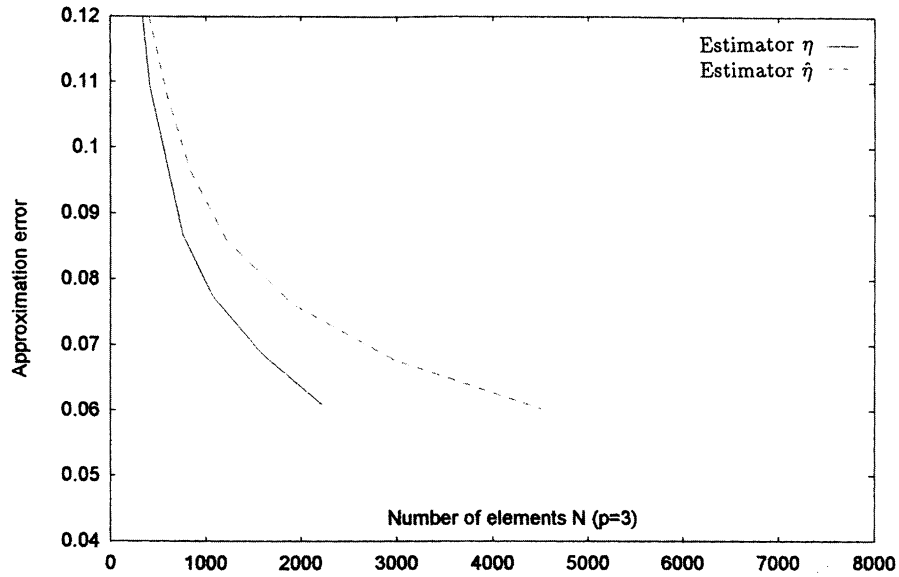


FIG. 3. *Error plots for the solution in* (4.7) ($p = 3$) *and the meshes generated by the error estimates* η *and* $\hat{\eta}$.

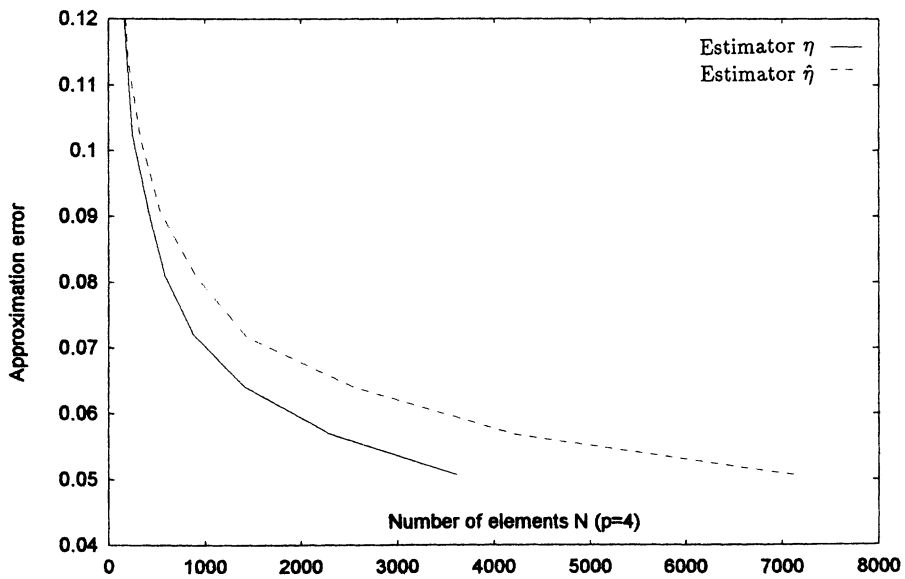


FIG. 4. *Error plots for the solution in* (4.7) ($p = 4$) *and the meshes generated by the error estimates* η *and* $\hat{\eta}$.

refinement guided by the error indicators based on the a posteriori error estimates η and $\hat{\eta}$. We follow the implementation strategy described in [28] though we adopted the average of the indicators as refinement criterion (see [25] for the details). The approximation error is computed using $\|u - u_h\|_{W^{1,p}}$, where u is the exact solution

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and u_h is the finite element approximation of u. The results may be summarized as follows. The performance of mesh refinement based on both a posteriori error estimates has been found to be very similar, when the solution is smooth, to that of the uniform refinement scheme. When the solution has a weaker singularity as in Example 4.1, where the first order derivative of the solution is continuous, the performance of the quasi-norm estimator η is just slightly (10%-20%) better than that of $\hat{\eta}$, as indicated in Figure 1 ($\sigma = 0.8$) and Figure 2 ($\sigma = 0.97$). However, when the solution has a stronger singularity, for instance, when the gradient is unbounded, it seems that the performance of the quasi-norm based refinement scheme is much better than that of the refinement scheme based on $\hat{\eta}$. We report here two test results of such cases for the following equation:

$$-\mathrm{div}(|\nabla u|^{p-2}\nabla u) = 0 \quad \text{in } (0,1)\times(0,1) \tag{4.7}$$

with the solution $u(x,y) = (x^2+y^2)^{(p-2)/2(p-1)}$. The results are included in Figure 3 ($p = 3$) and Figure 4 ($p = 4$). On these two examples, the quasi-norm based scheme only needs half of the number of the finite elements used in $\hat{\eta}$ scheme to achieve the same accuracy. Similar outcomes have also been found for other value of $p > 2$ for the equation.

5. Appendix. In the following we prove some inequalities which play an essential role in the establishment of our a posteriori error estimates. In all of these lemmas, C is a positive constant which only depends on p. The first two lemmas have been used in our work on a priori quasi-norm error bounds for the finite element approximation of degenerate nonlinear PDEs; see, e.g., [7] and [19]. The proof of the inequalities in Lemma 5.1 can be found in [9].

LEMMA 5.1. *For all $p > 1$, $\delta \geq 0$, ξ, $\eta \in R^n$,*

$$|\,|\xi|^{p-2}\xi - |\eta|^{p-2}\eta\,| \leq C|\xi-\eta|^{1-\delta}(|\xi|+|\eta|)^{p-2+\delta}, \tag{5.1}$$

$$(|\xi|^{p-2}\xi - |\eta|^{p-2}\eta, \xi-\eta) \geq C|\xi-\eta|^{2+\delta}(|\xi|+|\eta|)^{p-2-\delta}. \tag{5.2}$$

The following inequality has been proved in [9] for the case $\theta \in [1,\infty)$. We include here a complete proof as the case $\theta \in (0,1]$ is also important in this work.

LEMMA 5.2. *For all a, σ_1, $\sigma_2 \geq 0$, $p > 1$, $\theta > 0$,*

$$(a+\sigma_1)^{p-2}\sigma_1\sigma_2 \leq \theta^{-\gamma}(a+\sigma_1)^{p-2}\sigma_1^2 + \theta(a+\sigma_2)^{p-2}\sigma_2^2,$$

where

$$\gamma = \begin{cases} 1, & 1 < p \leq 2,\ \theta \in [1,\infty) \quad \text{or} \quad 2 < p < \infty,\ \theta \in (0,1), \\ \dfrac{1}{p-1}, & 1 < p \leq 2,\ \theta \in (0,1) \quad \text{or} \quad 2 < p < \infty,\ \theta \in [1,\infty). \end{cases}$$

Proof. The result clearly holds when $\sigma_1\sigma_2 = 0$. Thus, we assume $\sigma_1\sigma_2 \neq 0$. Let us consider the case $1 < p \leq 2$. First, suppose that $0 < \sigma_2 \leq \sigma_1$. Then, for any $\theta > 0$

$$(a+\sigma_1)^{p-2}\sigma_1\sigma_2 \quad \leq \quad \theta^{-1}(a+\sigma_1)^{p-2}\sigma_1^2 + \theta(a+\sigma_2)^{p-2}\sigma_2^2.$$

Further, if $\theta < 1$, then

$$(a+\sigma_1)^{p-2}\sigma_1\sigma_2 \leq \theta^{-\gamma}(a+\sigma_1)^{p-2}\sigma_1^2 + \theta(a+\sigma_2)^{p-2}\sigma_2^2,$$

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where we have noted the fact that $\theta^{-1} \le \theta^{-\frac{1}{p-1}} = \theta^{-\gamma}$, when $\theta < 1$ and $1 < p \le 2$.

Now suppose that $0 < \sigma_1 \le \sigma_2$. Then, for any $\theta > 0$

$$\begin{aligned}
&(\theta^{-\frac{1}{p-1}}(a+\sigma_1)^{p-2}\sigma_1^2 + \theta(a+\sigma_2)^{p-2}\sigma_2^2)/(a+\sigma_1)^{p-2}\sigma_1\sigma_2 \\
&= \theta^{-\frac{1}{p-1}}\frac{\sigma_1}{\sigma_2} + \theta\left(\frac{a+\sigma_1}{a+\sigma_2}\right)^{2-p}\frac{\sigma_2}{\sigma_1} \ge \theta^{-\frac{1}{p-1}}\frac{\sigma_1}{\sigma_2} + \theta\left(\frac{\sigma_1}{\sigma_2}\right)^{2-p-1} \\
&= \theta^{-\frac{1}{p-1}}\frac{\sigma_1}{\sigma_2} + \theta\left(\frac{\sigma_2}{\sigma_1}\right)^{p-1} = A^{-1} + A^{p-1} \ge 1,
\end{aligned}$$

where $A = \theta^{\frac{1}{p-1}}\frac{\sigma_2}{\sigma_1}$. We have used here the facts that $\frac{a+\sigma_1}{a+\sigma_2} \ge \frac{\sigma_1}{\sigma_2}$ if $0 < \sigma_1 \le \sigma_2$. Further, if $\theta \ge 1$, then $\theta^{-\frac{1}{p-1}} \le \theta^{-1} = \theta^{-\gamma}$. Therefore,

$$\begin{aligned}
(a+\sigma_1)^{p-2}\sigma_1\sigma_2 &\le \theta^{-\frac{1}{p-1}}(a+\sigma_1)^{p-2}\sigma_1^2 + \theta(a+\sigma_2)^{p-2}\sigma_2^2 \\
&\le \theta^{-\gamma}(a+\sigma_1)^{p-2}\sigma_1^2 + \theta(a+\sigma_2)^{p-2}\sigma_2^2.
\end{aligned}$$

Thus, we have proved Lemma 5.2 for $1 < p \le 2$. □

We can prove the results similarly for the case $p \ge 2$.

The following lemma is essential for estimating a bilinear form via the quasi norms. It can be viewed as a generalization of the Young inequality where a=0.

LEMMA 5.3. *For all $a, \sigma_1, \sigma_2 \ge 0$, $p > 1$, and $\delta > 0$,*

$$\sigma_1\sigma_2 \le \delta^{-\beta}(a^{p-1} + \sigma_1)^{p'-2}\sigma_1^2 + \delta(a+\sigma_2)^{p-2}\sigma_2^2,$$

where β is such that $\delta^{-\beta} = \max\{\delta^{-1}, \delta^{-\frac{1}{p-1}}\}$, and p' is such that $\frac{1}{p} + \frac{1}{p'} = 1$.

Proof. Without loss of generality, we assume $\sigma_1\sigma_2 \ne 0$. First, assume that $a > 0$. Then it follows from Lemma 5.2 that for any $\theta > 0$

$$\begin{aligned}
\sigma_1\sigma_2 &= (1+\sigma_1 a^{1-p})^{2-p}(1+\sigma_1 a^{1-p})^{p-2}\sigma_1\sigma_2 \\
&= (1+\sigma_1 a^{1-p})^{2-p}[(a+(\sigma_1 a^{2-p}))^{p-2}(\sigma_1 a^{2-p})\sigma_2] \qquad (5.3) \\
&\le (1+\sigma_1 a^{1-p})^{2-p}(\theta^{-\gamma}(a+\sigma_1 a^{2-p})^{p-2}a^{2(2-p)}\sigma_1^2 + \theta(a+\sigma_2)^{p-2}\sigma_2^2).
\end{aligned}$$

Note that $\gamma(2-p) \le \frac{2-p}{p-1}$ $\forall p > 1$, $\gamma = 1$ or $\frac{1}{p-1}$. Hence, $(1+\sigma_1 a^{1-p})^{\gamma(2-p)} \le (1+\sigma_1 a^{1-p})^{\frac{2-p}{p-1}}$ $\forall \sigma_1 \ge 0$, $a > 0$, since $(b^x)' \ge 0$ when $b \ge 1$. For any $\delta > 0$, let $\theta = \delta(1+\sigma_1 a^{1-p})^{p-2}$. Note $\frac{2-p}{p-1} = p' - 2$. It follows from (5.3) that

$$\begin{aligned}
\sigma_1\sigma_2 &\le \theta^{-\gamma}\sigma_1^2 a^{2-p} + \delta(a+\sigma_2)^{p-2}\sigma_2^2 \\
&= \delta^{-\gamma}(1+\sigma_1 a^{1-p})^{\gamma(2-p)}\sigma_1^2 a^{2-p} + \delta(a+\sigma_2)^{p-2}\sigma_2^2 \\
&\le \delta^{-\gamma}(1+\sigma_1 a^{1-p})^{\frac{2-p}{p-1}}\sigma_1^2 a^{2-p} + \delta(a+\sigma_2)^{p-2}\sigma_2^2 \\
&= \delta^{-\gamma}(a^{p-1}+\sigma_1)^{\frac{2-p}{p-1}}\sigma_1^2 + \delta(a+\sigma_2)^{p-2}\sigma_2^2 \\
&= \delta^{-\gamma}(a^{p-1}+\sigma_1)^{p'-2}\sigma_1^2 + \delta(a+\sigma_2)^{p-2}\sigma_2^2 \\
&\le \delta^{-\beta}(a^{p-1}+\sigma_1)^{p'-2}\sigma_1^2 + \delta(a+\sigma_2)^{p-2}\sigma_2^2.
\end{aligned}$$

Then, Lemma 5.3 follows when $a > 0$. □

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When $a = 0$, Lemma 5.3 follows from the standard Young inequality:

$$\sigma_1\sigma_2 \le \delta^{-\frac{1}{p-1}}\sigma_1^{p'} + \delta\sigma_2^p \le \delta^{-\beta}(a^{p-1}+\sigma_1)^{p'-2}\sigma_1^2 + \delta(a+\sigma_2)^{p-2}\sigma_2^2.$$

The following two lemmas will be used to prove some triangle-inequality–like results for a power of the quasi norms.

LEMMA 5.4. *For all* $1 < p < \infty$, $\sigma_1, \sigma_2 \in R^n$, *and* $a \ge 0$,

$$(a+|\sigma_1+\sigma_2|)^{p-2}|\sigma_1+\sigma_2|^2 \le C(a+|\sigma_1|)^{p-2}|\sigma_1|^2 + C(a+|\sigma_2|)^{p-2}|\sigma_2|^2.$$

Proof. Without loss of generality, we assume that $|\sigma_1|+|\sigma_2| > 0$. First, note that the function $x \to (a+x)^{p-2}x^2$ is increasing on $[0,\infty)$. Thus,

$$(a+|\sigma_1+\sigma_2|)^{p-2}|\sigma_1+\sigma_2|^2 \le (a+|\sigma_1|+|\sigma_2|)^{p-2}(|\sigma_1|+|\sigma_2|)^2. \tag{5.4}$$

When $1 < p \le 2$,

$$\begin{aligned}(a+|\sigma_1|+|\sigma_2|)^{p-2}(|\sigma_1|+|\sigma_2|)^2 &\le 2(a+|\sigma_1|+|\sigma_2|)^{p-2}(|\sigma_1|^2+|\sigma_2|^2)\\ &\le 2((a+|\sigma_1|)^{p-2}|\sigma_1|^2 + (a+|\sigma_2|)^{p-2}|\sigma_2|^2)\ .\end{aligned} \tag{5.5}$$

When $p \ge 2$,

$$\begin{aligned}&\frac{(a+|\sigma_1|)^{p-2}|\sigma_1|^2 + (a+|\sigma_2|)^{p-2}|\sigma_2|^2}{(a+|\sigma_1|+|\sigma_2|)^{p-2}(|\sigma_1|+|\sigma_2|)^2}\\ &= \left(\frac{a+|\sigma_1|}{a+|\sigma_1|+|\sigma_2|}\right)^{p-2}\left(\frac{|\sigma_1|}{|\sigma_1|+|\sigma_2|}\right)^2 + \left(\frac{a+|\sigma_2|}{a+|\sigma_1|+|\sigma_2|}\right)^{p-2}\left(\frac{|\sigma_2|}{|\sigma_1|+|\sigma_2|}\right)^2\\ &\ge \left(\frac{|\sigma_1|}{|\sigma_1|+|\sigma_2|}\right)^{p-2+2} + \left(\frac{|\sigma_2|}{|\sigma_1|+|\sigma_2|}\right)^{p-2+2} = \frac{|\sigma_1|^p+|\sigma_2|^p}{(|\sigma_1|+|\sigma_2|)^p} \ge C(p).\end{aligned}$$

Thus, $\forall p > 1$ there is a $C = C(p) > 0$ such that

$$\begin{aligned}(a+|\sigma_1|+|\sigma_2|)^{p-2}(|\sigma_1|+|\sigma_2|)^2 &\le C(a+|\sigma_1|)^{p-2}|\sigma_1|^2\\ &\quad + C(a+|\sigma_2|)^{p-2}|\sigma_2|^2. \qquad \square\end{aligned} \tag{5.6}$$

LEMMA 5.5. *For all* $1 < p < \infty$ *and* $\sigma, \sigma_1, \sigma_2 \in R^n$,

$$(|\sigma_1|+|\sigma_2|)^{p-2}|\sigma_1-\sigma_2|^2 \le C(|\sigma|+|\sigma-\sigma_1|)^{p-2}|\sigma-\sigma_1|^2 + C(|\sigma|+|\sigma-\sigma_2|)^{p-2}|\sigma-\sigma_2|^2.$$

Proof. First, note that $(|a|+|b|)/2 \le |a|+|a-b| \le 2(|a|+|b|)$ for $a, b \in R^n$. It follows from Lemma 5.1 that

$$\begin{aligned}&(|\sigma_1|+|\sigma_2|)^{p-2}|\sigma_1-\sigma_2|^2 \le C(|\sigma_1|^{p-2}\sigma_1 - |\sigma_2|^{p-2}\sigma_2, \sigma_1-\sigma_2)\\ &= C(|\sigma_1|^{p-2}\sigma_1 - |\sigma|^{p-2}\sigma + |\sigma|^{p-2}\sigma - |\sigma_2|^{p-2}\sigma_2, \sigma_1-\sigma+\sigma-\sigma_2)\\ &= C(|\sigma_1|^{p-2}\sigma_1 - |\sigma|^{p-2}\sigma, \sigma_1-\sigma) + C(|\sigma_1|^{p-2}\sigma_1 - |\sigma|^{p-2}\sigma, \sigma-\sigma_2)\\ &\quad + C(|\sigma|^{p-2}\sigma - |\sigma_2|^{p-2}\sigma_2, \sigma_1-\sigma) + C(|\sigma|^{p-2}\sigma - |\sigma_2|^{p-2}\sigma_2, \sigma-\sigma_2)\\ &\le C(|\sigma_1|+|\sigma|)^{p-2}|\sigma_1-\sigma|^2 + C(|\sigma_1|+|\sigma|)^{p-2}|\sigma_1-\sigma|\ |\sigma-\sigma_2|\\ &\quad + C(|\sigma_2|+|\sigma|)^{p-2}|\sigma-\sigma_2|\ |\sigma_1-\sigma| + C(|\sigma_2|+|\sigma|)^{p-2}|\sigma-\sigma_2|^2\ .\end{aligned} \tag{5.7}$$

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It follows that there is a $C = C(p) > 0$ such that $(i = 1, 2)$

$$(|\sigma| + |\sigma_i|)^{p-2} \leq C(|\sigma| + |\sigma - \sigma_i|)^{p-2}.$$

Then it follows from (5.7) and Lemma 5.2 that

$$(|\sigma_1| + |\sigma_2|)^{p-2}|\sigma_1 - \sigma_2|^2 \leq C(|\sigma| + |\sigma - \sigma_1|)^{p-2}|\sigma - \sigma_1|^2 + C(|\sigma| + |\sigma - \sigma_2|)^{p-2}|\sigma - \sigma_2|^2.$$

Then, Lemma 5.5 follows. □

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